



Sri Chaitanya IIT Academy.,India.

A.P. T.S. KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S) & LIIT

JEE-MAIN

Date: 28-12-2025

Time: 09.00Am to 12.00Pm

GTM-06

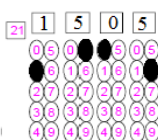
Max. Marks: 300

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
 2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
 3. The test is of **3 hours** duration.
 4. The Test Booklet consists of **75 Questions**. The maximum marks are **300**.
 5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics and Chemistry** having **25 Questions** in each subject and subject having **two sections**.
(I) Section-I contains **20 Multiple Choice Questions** with only one correct option.
Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
(II) Section-II contains **05 Numerical Value Type Questions**.
- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking



Question Cancelled for Marking

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

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RANK 1

341

RANK

1

JEE Advanced 2023

VAVILALA CHIVILAS REDDY

RANK 1

341

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MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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**MATHEMATICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- If $f(x)$ is a differentiable function such that $f'(1) = 4$ and $f'(4) = \frac{1}{2}$, then value of $\lim_{x \rightarrow 0} \frac{f(x^2 + x + 1) - f(1)}{f(x^4 - x^2 + 2x + 4) - f(4)}$ is
 1) 8 2) 16 3) 4 4) Does not exist
- If normal at point (h, k) on parabola $y^2 = 8x$ meets parabola again at point $(18, 12)$, then sum of all possible value of h , is
 1) 8 2) 12 3) 10 4) 6
- Let $(1, 0)$ be the vertex and $x + y - 7 = 0$ be the equation of the latus rectum of a parabola and if the equation of parabola is $x^2 + y^2 - 2xy + px + qy + r = 0$, then $(p + q + r)$ is equal to
 1) -28 2) -27 3) -23 4) -22
- The differential representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$, is a parameter, is of order and degree as follows
 1) order 1, degree 2 2) order 1, degree 1
 3) order 1, degree 3 4) order 2, degree 2
- Statement-1:** The number of integral terms in $(\sqrt{5} + \sqrt[8]{7})^{1024}$ is 129
Statement-2: The sum of rational terms in $(\sqrt{2} + \sqrt[3]{3} + \sqrt[6]{5})^{10}$ is 1262
Statement-3: The number of irrational terms in the expansion of $(5^{1/6} + 2^{1/8})^{100}$ is 97
 The correct statements among the following are
 1) 1,2 2) 2,3 3) 1,3 4) 1,2&3

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6. If $g(x) = x^2 + \frac{1}{x^2}$ and $f: [2, \infty) \rightarrow (-\infty, \infty)$, such that $f(g(x)) = x^6 + \frac{1}{x^6}$, then $f'(x)$ equals
- 1) $x^5 + \frac{1}{x^5}$ 2) $3x^2 - 3$ 3) $5x^5 - \frac{6}{x^7}$ 4) $5x^5 - \frac{5}{x^5}$
7. Number of points at which the function $f(x) = |\cos|x|| + \cos^{-1}(\operatorname{sgn} x) + |\ln x|$ is not differentiable in $(0, 2\pi)$ is
- 1) 1 2) 2 3) 3 4) 4
8. If the locus of the middle point of chords of an ellipse $\frac{x^2}{3} + \frac{y^2}{4} = 1$ passing through $(2, 0)$ is another ellipse A, then the length of latus rectum of the ellipse A is
- 1) $\frac{8}{3}$ 2) $\sqrt{3}$ 3) $\frac{1}{\sqrt{3}}$ 4) $\frac{3}{8}$
9. Number of different words that can be formed from all letters of word **APPLICATION** such that two vowels never come together is
- 1) $(45)7!$ 2) $8!$ 3) $6!7!$ 4) $(32)6!$
10. If $\sum_{i=1}^{10} (x_i - 5) = 5$ and $\sum_{i=1}^{10} (x_i - 5)^2 = 125$, variance of $x_1, x_2, \dots, x_{10}$ will be
- 1) $\frac{9}{4}$ 2) $\frac{25}{4}$ 3) $\frac{49}{4}$ 4) $\frac{81}{4}$
11. If probability that $2^3 3^5 5^4 7^3 11^\alpha$ be resolved as product of two factors such that their highest common factor is 1 or 5 or 7 is $\left(\frac{1}{35}\right)$, then 3α is
- 1) 3 2) 6 3) 15 4) 18





12. If $\begin{vmatrix} x & 1 & 1 \\ 2 & x & 3 \\ x & -1 & 2 \end{vmatrix} = 4$ (where x is a complex number), then the value of the determinant

$$\begin{vmatrix} 2x+3 & -3 & 3-x \\ 3x-4 & x & 2-3x \\ -x^2-2 & 2x & x^2-2 \end{vmatrix}$$
 is equal to

- 1) 64 2) 16 3) 8 4) 10

13. Let $x_1 = \sum_{r=0}^{\infty} \left(\frac{\sin r \theta \cos^r \theta}{r!} \right)$ and $x_2 = \sum_{r=0}^{\infty} \frac{\cos r \theta \cos^r \theta}{r!}$, then which of the following is

INCORRECT $\left(0 < \theta < \frac{\pi}{2} \right)$?

- 1) $x_1^2 + x_2^2 = e^{2\cos^2 \theta}$ 2) $x_1 + x_2 = e^{\cos^2 \theta} \sin \left(\frac{\pi}{4} + \sin \theta \cos \theta \right)$
 3) $x_2^2 - x_1^2 = e^{2\cos^2 \theta} \cos(2\sin \theta \cos \theta)$ 4) $x_1 x_2 = \frac{e^{2\cos^2 \theta}}{2} \sin(2\sin \theta \cos \theta)$

14. The shortest distance between the y-axis and the line

$$2x + 3y + 5z + 1 = 3x + 4y + 6z + 2 = 0 \text{ is } \frac{2}{\sqrt{k}}, \text{ then the value of 'k' is}$$

- 1) 2 2) 3 3) 5 4) 6

15. If $\frac{1}{1^5} + \frac{1}{2^5} + \frac{1}{3^5} + \frac{1}{4^5} + \dots$ upto infinite terms = k , then value of

$\left(\frac{1}{1^5} + \frac{1}{3^5} + \frac{1}{5^5} + \frac{1}{7^5} + \dots \right)$ upto infinite terms will be

- 1) $\frac{k}{32}$ 2) $\frac{31k}{32}$ 3) $\frac{k}{16}$ 4) $\frac{15k}{16}$

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16. The area bounded between $x^2 - y = 1$ and $y = |x - 1| + x$ is
- 1) $\frac{7 + 4\sqrt{2}}{3}$ 2) $\frac{7 - 4\sqrt{2}}{3}$ 3) $\frac{5 + 4\sqrt{2}}{3}$ 4) $\frac{8 + 4\sqrt{2}}{3}$
17. A variable circle which always touches the line $x + y - 2 = 0$ at point $(1, 1)$ cuts circle $x^2 + y^2 + 4x + 5y - 4 = 0$ at P and Q, then line PQ always passes through fixed point
- 1) $(1, 1)$ 2) $(4, -2)$ 3) $(8, -6)$ 4) $(0, 0)$
18. In a triangle ABC, co-ordinates of A are $(1, 2)$ and the equations to the medians through B and C are $x + y = 5$ and $x = 4$ respectively. Then the co-ordinates of B and C will be
- 1) $(-2, 7), (4, 3)$ 2) $(7, -2), (4, 3)$ 3) $(2, 7), (-4, 3)$ 4) $(2, -7), (3, -4)$
19. **Statement-1:** If A is square matrix of order 3 such that $|A| = 3$, then $|\text{adj}(\text{adj}(\text{adj}A))|$ is 3^8
- Statement-2:** If A and B are square matrices of order 3×3 , where $|A| = 2$ and $|B| = 3$, then $\left| (A^{-1}) \cdot \text{adj}(B^{-1}) \cdot \text{adj}(2A^{-1}) \right| = \frac{8}{9}$
- Statement-3:** A is a non-singular matrix of order $n \times n$ such that $3ABA^{-1} + A = 2A^{-1}BA$ then $|A + B| = 0$
- The correct statements among the following are
- 1) 1,2 2) 2,3 3) 1,3 4) 1,2&3
20. The value of $\int_0^{\sin^2 x} \sin^{-1}(\sqrt{t}) dt + \int_0^{\cos^2 x} \cos^{-1}(\sqrt{t}) dt$ is equal to _____
- 1) 1 2) $\frac{\pi}{4}$ 3) 0 4) $-\pi$





SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Let A, B, C, D be the vertices of a regular tetrahedron each of whose edges measures 1 meter. A bug, starting from vertex A, observes, the following rules, at each vertex it chooses one of the three edges meeting at that vertex, each edge being equally likely to be chosen, and crawls along that edge to the vertex at its opposite end. Let $p = \frac{n}{729}$ be the probability that the bug is at vertex A when it has crawled exactly 7 meters. The value of n is
22. Let $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + \hat{k}$ and vector $\vec{c}$ satisfying conditions (i) $[\vec{a} \ \vec{b} \ \vec{c}] = 0$
(ii) $\vec{b} \cdot \vec{c} = 0$ (iii) $\vec{a} \cdot \vec{c} = 7$. Then the value of $\frac{2}{7}|\vec{c}|^2$ is equal to
23. If slope of the chord of hyperbola $2x^2 - 3y^2 = 6$ which is bisected at (5, 3) is $\frac{p}{q}$, where p and q are co-prime positive integers, then the value of $|p - q|$ is
24. The number of solutions of equation $9\cos^{12}x + \cos^2 2x + 1 = 6\cos^6 x \cos 2x + 6\cos^6 x - 2\cos 2x$ in $[0, 4\pi]$ are
25. Normal at B to the circle $x^2 + y^2 = 5$ intersects it again at C. If a line through C intersecting the circle at D meet the tangent to it at B at A(3, 1) then length of CD is





PHYSICS

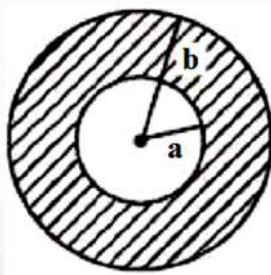
Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. A circular disc of radius b has a hole of radius a at its centre (see figure). If the mass per unit area of the disc varies as $\left(\frac{\sigma_0}{r}\right)$, where r is distance from axis, then the radius of gyration of the disc about its axis passing through the centre is



- 1) $\frac{a+b}{3}$ 2) $\sqrt{\frac{a^2 + b^2 + ab}{3}}$ 3) $\frac{a+b}{2}$ 4) $\sqrt{\frac{a^2 + b^2 + ab}{2}}$
27. An excited He^+ ion emits two photons in succession, with wavelengths 108.5 nm and 30.4 nm, in making a transition to ground state. The quantum number n , corresponding to its initial excited state is (for photon of wavelength λ , energy $E = \frac{1240eV}{\lambda(\text{in nm})}$)
- 1) $n = 4$ 2) $n = 6$ 3) $n = 5$ 4) $n = 7$
28. A cylinder has inner radius r_1 and outer radius r_2 . The inner face has higher temperature than the outer surface. Heat is flowing through the cylinder radially. In steady state, the rate of heat flow is inversely proportional to (the temperature of inner and outer surfaces are maintained constant)

- 1) $\frac{r_1 r_2}{r_2 - r_1}$ 2) $\log\left(\frac{r_2}{r_1}\right)$ 3) $\frac{r_1^2}{r_1 + r_2}$ 4) $r_2 - r_1$

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29. A circular coil of one turn is made into a square loop of two turns. If same current is passing through the coils, then the magnetic moments of coils are M_1 and M_2

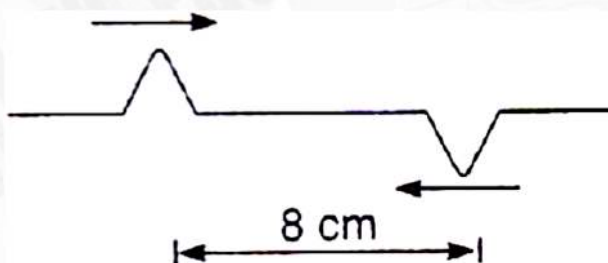
respectively. $\frac{M_1}{M_2}$ is

- 1) $\frac{8}{\pi}$ 2) $\frac{\pi}{2}$ 3) $\frac{4}{\pi^2}$ 4) $\frac{3}{\pi}$

30. A particle executes SHM according to relation $y = 2 \cos\left(\frac{\pi}{6}t + \frac{\pi}{3}\right)m$. The distance travelled by the particle from time $t_1 = 1\text{sec}$ to $t_2 = 7\text{sec}$ is

- 1) 2 m 2) 4 m 3) 1 m 4) 3 m

31. Two identical but inverted pulses in a stretched string, whose centers are initially 8 cm apart, are moving towards each other at $t = 0$ as shown in the figure. The speed of each pulse is 2 cm/s. At time $t = 2\text{s}$ the total energy of the pulses will be



- 1) Zero 2) Purely kinetic
3) Purely potential 4) Partly kinetic and partly potential
32. In a double slit experiment, when a thin film of thickness t having refractive index μ is introduced in front of one of the slits, the maximum at the centre of the fringe pattern shifts by one fringe width. The value of t is (λ is the wavelength of the light used):
- 1) $\frac{\lambda}{2(\mu-1)}$ 2) $\frac{\lambda}{(\mu-1)}$ 3) $\frac{\lambda}{(2\mu-1)}$ 4) $\frac{2\lambda}{(\mu-1)}$





33. Given below are statements:

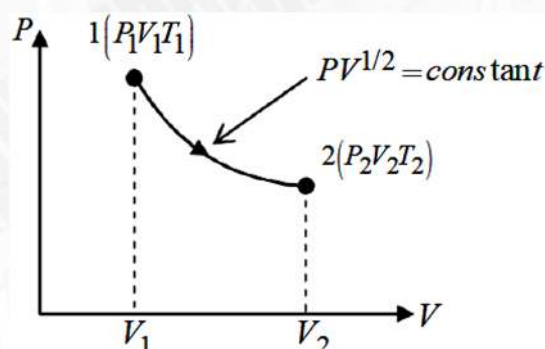
Statement-I: Two photons having equal linear momenta have equal wavelengths.

Statement-II: If the wavelength of photon is decreased, then the momentum and energy of a photon will also decrease.

In the light of the above statements, choose the correct answer from the options given below.

- 1) Statement-I is false but Statement-II is true.
- 2) Both Statement-I and Statement-II are true.
- 3) Both Statement-I and Statement-II are false.
- 4) Statement-I is true but Statement-II is false.

34. Thermodynamic process is shown below on a $P-V$ diagram for one mole of an ideal gas. If $V_2 = 2V_1$ then the ratio of temperatures T_2 / T_1 is



- 1) $\frac{1}{\sqrt{2}}$
- 2) $\frac{1}{2}$
- 3) 2
- 4) $\sqrt{2}$

35. A point dipole $\vec{p} = -p_0 \hat{x}$ is kept at the origin. The potential and electric field due to this dipole on the y -axis at a distance d are, respectively: (Take $V = 0$ at infinity)

- 1) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$
- 2) $0, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$
- 3) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$
- 4) $0, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$



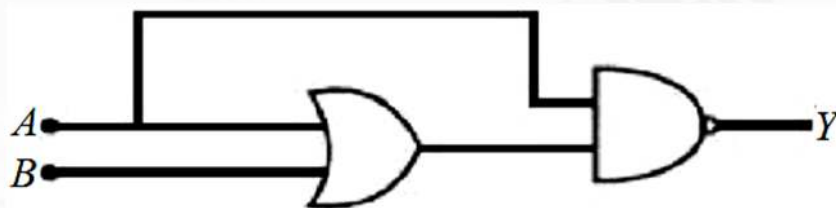


36. An electromagnetic wave is represented by the electric field $\vec{E} = E_0 \hat{n} \sin[\omega t + (6y - 8z)]$.

Taking unit vectors in x, y and z directions to be $\hat{i}, \hat{j}, \hat{k}$, the direction of propagation $\hat{s}$, is

- 1) $\hat{s} = \left(\frac{-3\hat{j} + 4\hat{k}}{5} \right)$ 2) $\hat{s} = \left(\frac{4\hat{j} - 3\hat{k}}{5} \right)$
 3) $\hat{s} = \left(\frac{-4\hat{k} + 3\hat{j}}{5} \right)$ 4) $\hat{s} = \left(\frac{-3\hat{i} - 4\hat{j}}{5} \right)$

37. The truth table for the circuit given in the figure is:



- 1)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

 2)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

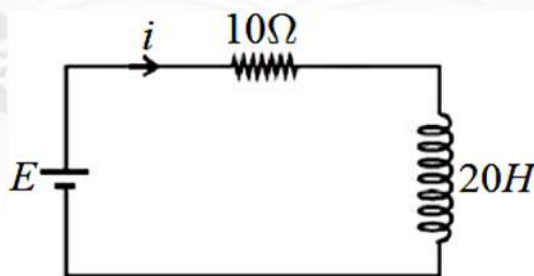
 3)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

 4)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

38. A 20 Henry inductor coil is connected to a 10 ohm resistance in series as shown in figure. The time (in s) at which rate of dissipation of energy (Joule's heat) across resistance is equal to the rate at which magnetic energy is stored in the inductor, is (current at $t = 0$ is zero)



- 1) $\frac{2}{\ln 2}$ 2) $\ln 2$ 3) $\frac{1}{2} \ln 2$ 4) $2 \ln 2$





39. A glass rod of radius 1 mm is inserted symmetrically into a glass capillary tube with inside radius 2 mm. Then the whole arrangement is brought in contact with the surface of water. Surface tension of water is $7 \times 10^{-2} \text{ Nm}^{-1}$. To what height will the water rise in the capillary? ($\theta = 0^\circ$)

1) 1.4 cm 2) 4.2 cm 3) 2.1 cm 4) 6.8 cm

40. A rod of length ℓ , mass M and cross sectional area A is placed on a rough horizontal surface. A horizontal force F is applied to rod as shown in figure. The coefficient of friction between rod and surface is μ , the Young's modulus of material of rod is Y . (Assume that friction is distributed uniformly on rod)



The elongation in rod if $F > \mu Mg$ is

1) $\frac{(F - \mu Mg)\ell}{2AY}$ 2) $\left[\frac{F - \frac{\mu Mg}{2}}{2AY} \right] \ell$ 3) $\frac{F\ell}{2AY}$ 4) $\frac{F\ell}{4AY}$

41. The ratio of mass densities of nuclei of ^{40}C and ^{16}O is close to
- 1) 0.1 2) 5 3) 2 4) 1

42. Given below are two statements:

Statement-I: A second's pendulum has a time period of 1 second.

Statement-II: Second's pendulum takes precisely one second to move between the two extreme positions of its oscillation.

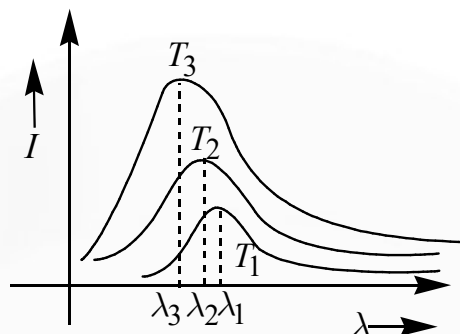
In the light of the above statements, choose the correct answer from the options given below.

- 1) Both Statement-I and Statement-II are false.
 2) Statement-I is true but Statement-II is false.
 3) Statement-I is false but Statement-II is true.
 4) Both Statement-I and Statement-II is true.





43. According to Wien's displacement law.



A = Area under the curve

I = Intensity of radiation

- 1) Total area under the three graphs A_1, A_2, A_3 are equal
 - 2) Areas are in the ratio: $A_1 : A_2 : A_3 = T_1^2 : T_2^2 : T_3^2$
 - 3) Areas are in the ratio: $A_1 : A_2 : A_3 = T_1^4 : T_2^4 : T_3^4$
 - 4) $T_3 < T_2 < T_1$
44. A circuit connected to an ac source of emf $e = e_0 \sin(1000t)$ with t in seconds, gives a phase difference of $\frac{\pi}{4}$ between the emf e and current i . Which of the following circuits will exhibit this?
- 1) RC circuit with $R = 1 \text{ k}\Omega$ and $C = 10 \mu\text{F}$
 - 2) RL circuit with $R = 1 \text{ k}\Omega$ and $L = 10 \text{ mH}$
 - 3) RL circuit with $R = 1 \text{ k}\Omega$ and $L = 1 \text{ mH}$
 - 4) RC circuit with $R = 1 \text{ k}\Omega$ and $C = 1 \mu\text{F}$
45. If ' M ' is the mass of water that rises in a capillary tube of radius ' r ', then mass of water which will rise in a capillary tube of radius ' $2r$ ' is
- 1) $4M$
 - 2) $\frac{M}{2}$
 - 3) M
 - 4) $2M$





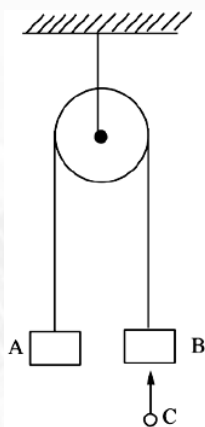
SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. Two identical blocks A and B each of mass 2 kg are hanging stationary by a light inextensible flexible string, passing over a light and frictionless pulley, as shown in figure. A particle C of mass 1 kg moving vertically upwards with velocity 18ms^{-1} collides with block B and gets stuck to it. The time after which the string becomes taut again is $(T/10)\text{s}$. Then T is (Take, $g = 10\text{ms}^{-2}$)



47. A cylindrical wire of radius 0.5 mm and conductivity $5 \times 10^7 \text{ S/m}$ is subjected to an electric field of 10 mV/m. The value of current in the wire will be $x^3 \pi \text{ mA}$. The value of x is _____
48. The percentage increase in the speed of transverse waves produced in a stretched string if the tension is increased by 4% will be _____ %.
49. The root mean square speed of molecules of a given mass of a gas at 27°C and 1 atmosphere pressure is 200ms^{-1} . The root mean square speed of molecules of the same gas at 127°C and 2 atmosphere pressure is $\frac{x}{\sqrt{3}}\text{ms}^{-1}$. The value of x will be _____
50. The initial velocity v_i required to project a body vertically upward from the surface of the earth to reach a height of $10R$, where R is radius of the earth, may be described in terms of escape velocity v_e such that $v_i = \sqrt{\frac{x}{11}} \times v_e$. The value of x will be _____

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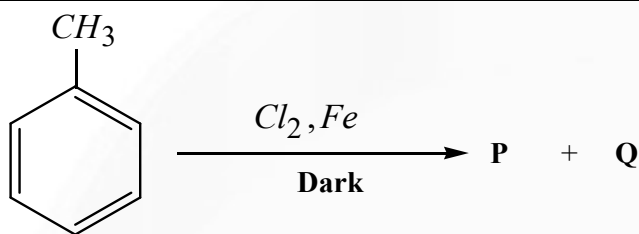
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**CHEMISTRY****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51.



Select the correct set of statements from the following list

- I. **P & Q** are easily separated due to large difference in melting points.
- II. **P & Q** are positional isomers.
- III. Locant number of halogen in IUPAC name of **P** and **Q** can be same
- IV. **P & Q** are not positional isomers as locate number of ' CH_3 ' group is different in their IUPAC name

- 1) II only 2) I, II, III only 3) I, III, IV only 4) II, III only

52. Assertion(A): Fluorine can stabilize highest possible oxidation state of chromium in the corresponding fluoride.

Reason(R): CrF_6 is covalent with higher bond enthalpy.

Considering the above statements, select the correct option.

- 1) '**A**' is correct and '**R**' is incorrect.
- 2) '**A**' is incorrect and '**R**' is correct.
- 3) Both **A & R** are correct and '**R**' is correct explanation of **A**.
- 4) Both **A & R** are correct but '**R**' is not correct explanation of **A**.



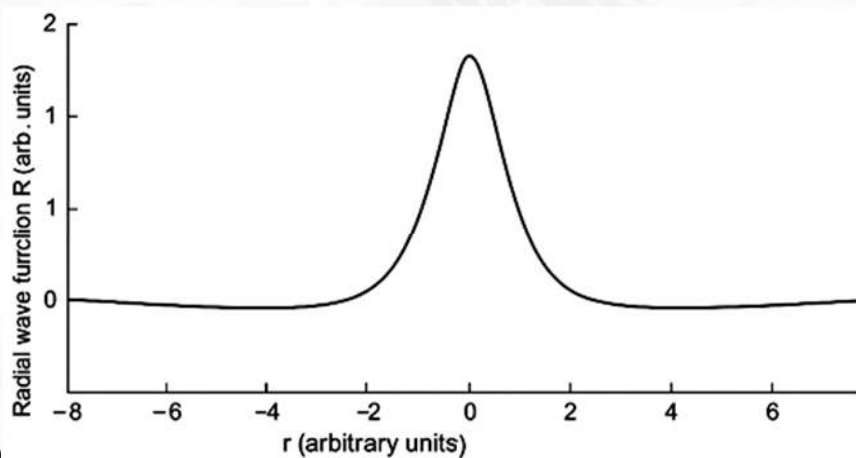


53. During photosynthesis, the balanced detailed equation is often written as

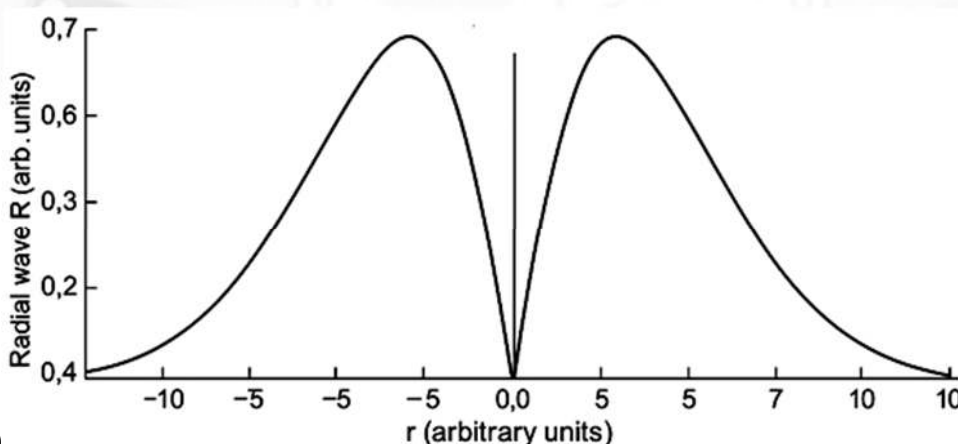
$6CO_2 + 12H_2O \xrightarrow{h\nu} C_6H_{12}O_6 + 6H_2O + 6O_2$. Which of the following correctly explains the reason for using 12 H_2O molecules on reactant side and '6' H_2O molecules in product side?

- 1) Six water molecules are consumed in the products side and six water molecules are formed in reactants side.
- 2) Twelve water molecules undergo photolysis to supply electrons and protons for CO_2 reduction, where as six water molecules are regenerated during glucose synthesis
- 3) Six water molecules are consumed in the reactant side and six water molecules are formed in product side.
- 4) Twelve water molecules are needed because O_2 comes from CO_2 and the excess six water molecules are left unreacted after glucose formation

54. Which of the following could be a possible qualitative radial wave function of hydrogen atomic orbitals with Azimuthal quantum number as 1?



1)



2)

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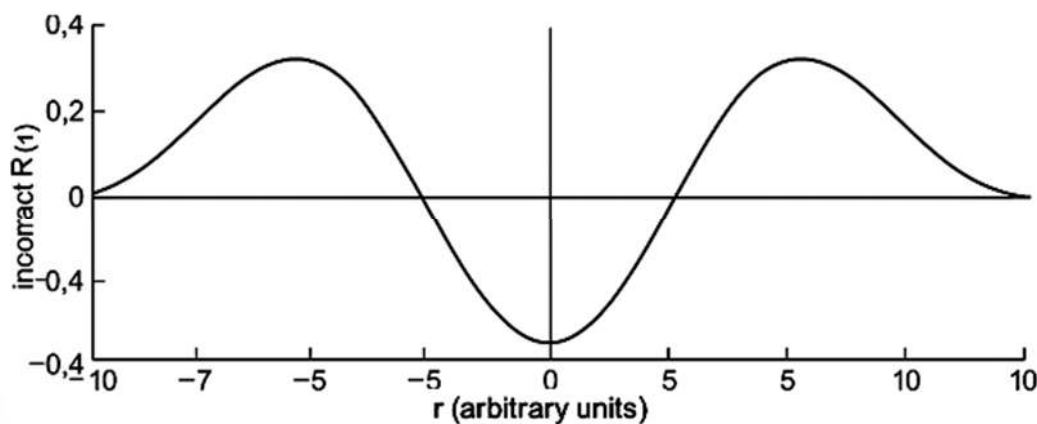
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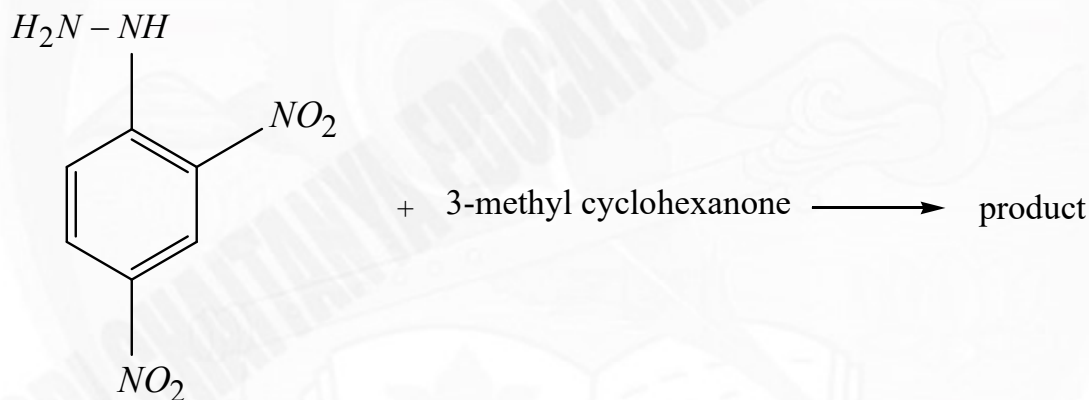
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3)

4) All are possible

55. Number of possible stereo isomers in the product of following reaction is



1) 3

2) 4

3) 8

4) 1

56. Which of the following dynamic equilibrium are not significantly affected by pressure?

I. $A_{(s)} + H_2O \rightleftharpoons A_{aq}$: here $A_{(s)}$ & H_2O are highly incompressible

II. $A_{(g)} + H_2O \rightleftharpoons A_{aq}$

III. $2A_{(g)} + H_2O \rightleftharpoons A_{2(g)}$

IV. $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$

1) I, IV only

2) II, III only

3) IV only

4) I only

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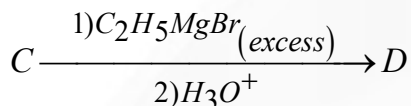
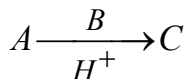
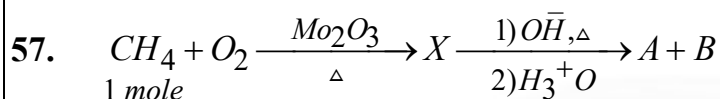
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1



A gives positive tollen's test.

Select correct statement if all reaction are 100% efficient

- 1) Number of moles of **D** formed is 1
 - 2) **D** given positive iodoform test
 - 3) At least one of the reaction involve formation of 2° alcohol
 - 4) **A** and **D** are members of Homologues
58. Statement-I: A reversible process proceeds infinitely slowly by a series of equilibrium states such that system and surroundings are always near equilibrium with each other
Statement-II: Gas expansion in an isolated system is irreversible due to system and surroundings being not in contact.
- In the view of above statements select the correct option
- 1) Both Statement-I and Statement-II are correct and Statement-I is correct reason of Statement-II
 - 2) Both Statement-I and Statement-II are correct and Statement-I is not correct explanation of Statement-II
 - 3) Statement-I is correct Statement-II is incorrect
 - 4) Statement-I is incorrect Statement-II is correct
59. Which of the following apparatus does not effect titre reading though contains small amount of distilled water prior to preparation of standard solution of analyte and during titration?
- I. Burette II. Pipette III. Conical flask IV. Standard flask
- 1) II, III, IV only 2) I, II, III, IV 3) III, IV only 4) I, II only



60. Which of the following reactions result the product with retention of configuration when react with HCl?

- 1) 2-pentanol 2) 3-methyloctan-4-ol
3) 5-methylheptan-1-ol 4) butanol

61. Match the following.

Column-I (Quantities)		Column-II (Factors effecting)	
P)	Molar conductance ($\wedge_m$)	A)	Temperature
Q)	Emf of cell with spontaneous cell reaction	B)	Concentration of chemical species involved in cell reaction.
R)	Electrode potential	C)	Nature of electrolyte
S)	Standard reduction potential	D)	Dilution

- 1) P-AB; Q-ABCD; R-ABD; S-C
- 2) P-AB; Q-ABC; R-ABCD; S-A
- 3) P-ABCD; Q-ABCD; R-ABCD; S-AC
- 4) P-ACD; Q-ABD; R-ABCD; S-C

62. $A + HIO_4 \longrightarrow 3HCOOH + HCHO + CHO - COOH$

Select 'A' where 'A' is a derivative of glucose.

- 1) Glucuronic acid 2) Glucaric acid 3) Glucitol 4) Gluconic acid

63. Assertion: Spin only magnetic moment of $[MnBr_4]^{2-}$ is 5.9 BM.

Reason: one 4s orbital and three 4p orbitals undergo hybridization in Mn^{+2} ion.

Select the correct statements.

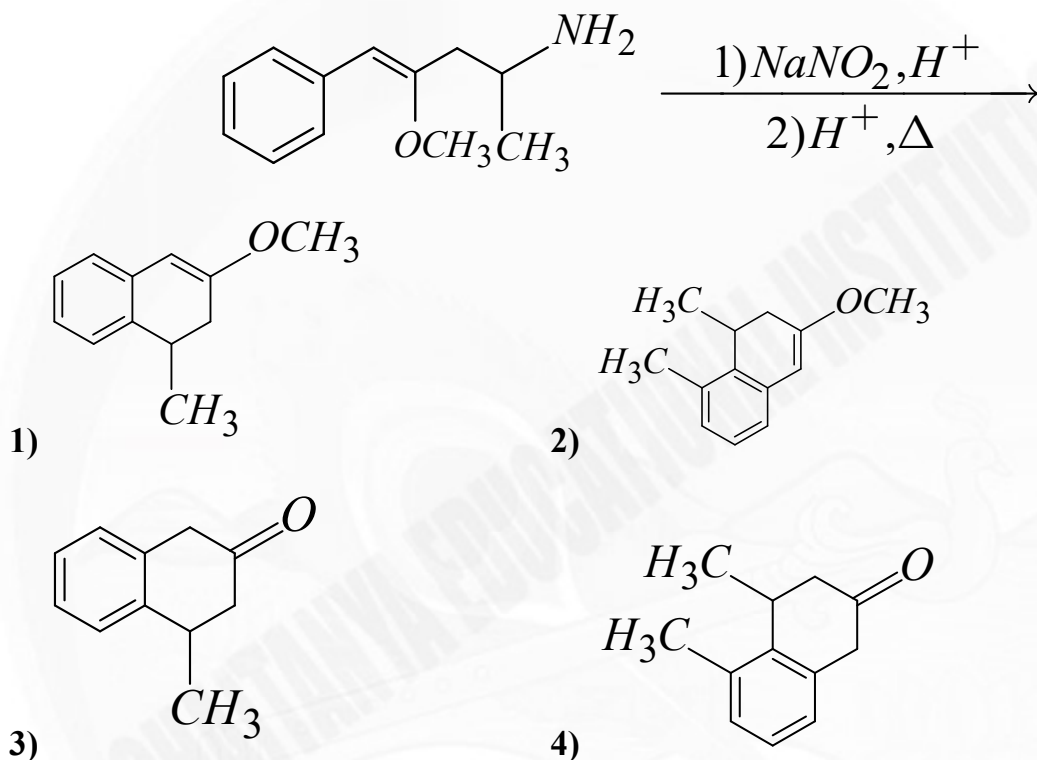
- 1) Assertion is correct and Reason is incorrect.
- 2) Both Assertion and Reason are correct and Reason is correct explanation of Assertion.
- 3) Both Assertion and Reason are incorrect
- 4) Assertion is correct and Reason is correct but Reason is not correct explanation of Assertion.



64. If solubility of Hg_2Cl_2 is 'x' then its solubility product is

- 1) $4x^2$ 2) $4x^3$ 3) x^3 4) $4x^4$

65. The major product of the following reaction is



66. Correct statements is/are

- i) HNO_2 undergoes disproportionation to form NO and HNO_3 .
 ii) Nitrogen shows maximum covalency is 5.
 iii) Bismuth pentahalides are highly stable except BiF_5 .
 iv) Phosphorous maximum covalency is 5.

- 1) i, ii only 2) iii, iv only 3) i, ii, iii, iv 4) i only

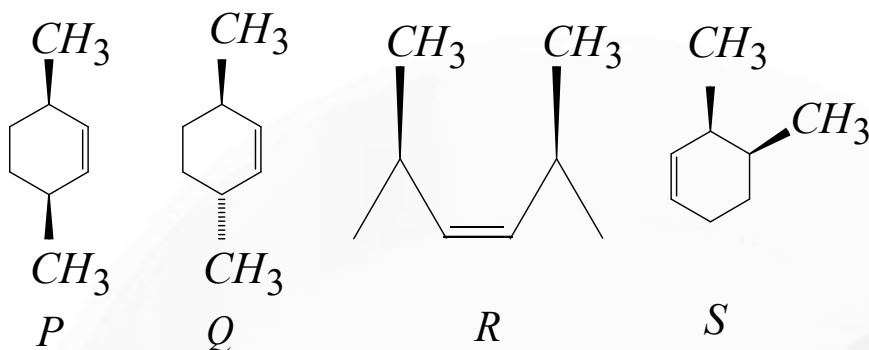
67. If relative decrease in vapour pressure of 3 moles of water when 1 mol of $NaCl$ is added is 0.4. What is the fraction of undissociated $NaCl$?

- 1) 60% 2) 20% 3) 100% 4) 0





68. Which of the following set gives meso product on reductive ozonolysis.



- 1) PQ only 2) P only 3) QR only 4) PR only

69. Polar oxy-fluoride of an element 'X' is

- 1) XeO_3 2) $XeOF_4$ 3) SF_4 4) Both 2 and 3

70. A cation of a salt solution produces a chocolate-brown precipitate with alkaline

$K_4[Fe(CN)_6]$. What is the colour of that aqueous ion solution?

- 1) Yellow 2) Blue 3) Green 4) Red

SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

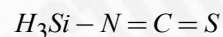
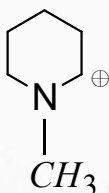
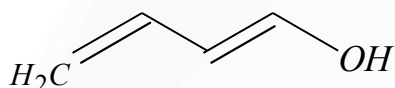
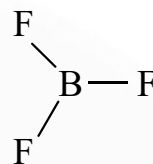
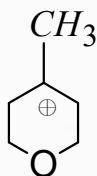
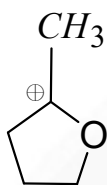
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. The rate constant of a reaction at $27^\circ C$ is $3 \times 10^{-3} \text{ min}^{-1}$. The fraction of molecules crossing the energy barrier of the reaction at that temperature is $10^{-3}\%$. If temperature is increased, the maximum value of rate constant k can be attained is $x \times 100 \text{ min}^{-1}$. What is the value of x ?
72. In an atom, the total number of electrons having quantum numbers $n = 4, |m_l| = 1$ and $m_s = -1/2$ is _____

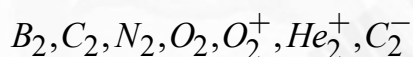




73. How many the following have double bond character between least electronegative (exclude hydrogen) and most electronegative atom in the list given below?



74. The number of species with fractional bond order is x and the number of paramagnetic species are y . The value of $|x - y|$ is _____



75. The enthalpy of formation of $C_2H_5OH(l)$ is -66 kcal/mol. The enthalpy of combustion of $CH_3OCH_3(g)$ is -348 kcal/mol. Enthalpy of formation of $H_2O(l)$ and $CO_2(g)$ are -68 kcal/mol and -94 kcal/mol respectively. Then the enthalpy of isomerisation of $C_2H_5OH(l)$ into $CH_3OCH_3(g)$ is $11x$ kcal/mol. The value of ' x ' is _____





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